

SUCHAKRA KUMAR GATTU

[Github](#) — [LinkedIn](#) — [LeetCode](#) — suchakrakumargattu@gmail.com
[+91 8555059710](#) — [Hyderabad, Telangana](#)

Summary

Software Engineer specializing in Speech AI, telephony-grade audio pipelines, and LLM inference optimization. Experienced in building and shipping production Speech-to-Text (STT) and Text-to-Speech (TTS) pipelines for real-time voice platforms, tuning streaming inference for low latency, and building evaluation frameworks for model quality and regression testing.

Skills

Speech & Voice AI: STT/ASR, TTS, SIP, Telephony, Streaming Audio Inference, Voice Agents, Real-time Conversational Pipelines

AI / Machine Learning: PyTorch, TensorFlow, Transformers, LoRA, QLoRA, SFT, DPO, PEFT, RAG, LangChain, LangGraph

Model Optimization & Evaluation: ONNX, TensorRT, llama.cpp, Quantization, Inference Scheduling (Dynamo), Offline Evaluation Pipelines, Hallucination Detection, Answer Relevance & Context Scoring

Programming Languages: Python, Java, JavaScript, C, TypeScript

Backend: FastAPI, Flask, Django, Node.js, Express.js, SQLAlchemy, Alembic, Prisma

Frontend: React Frameworks: Tanstack Router, Tanstack Query, React-router-dom

Databases: PostgreSQL, MySQL, MongoDB

MLOps / Infrastructure: Docker, MLflow, Prometheus, Grafana, cAdvisor, Node Exporter

Experience

Software Engineer

May 2026 – Present

CloudAngles, Hyderabad

- Engineered the complete telephony-grade audio pipeline for an enterprise Voice AI platform, integrating SIP trunking, call routing, Speech-to-Text (STT), LLM inference, and Text-to-Speech (TTS) into a production-ready streaming system.
- Designed and implemented scalable SIP workflows supporting reliable inbound and outbound AI voice calls, handling real-time telephony audio end to end.
- Reduced end-to-end voice conversation latency to approximately **1.15 seconds** through streaming inference and pipeline optimization across the STT/LLM/TTS stack, significantly improving real-time conversational quality.
- Integrated **Dynamo** for self-hosted LLM inference, improving execution efficiency through optimized inference scheduling for low-latency streaming responses.
- Analyzed and optimized the LLM prefill and decode phases, refactoring the inference pipeline to improve throughput and reduce response latency in a real-time voice context.
- Collaborated with cross-functional engineering teams to deploy scalable AI infrastructure and production-ready voice automation services.

Software Engineer Trainee

Aug 2025 – Apr 2026

CloudAngles, Hyderabad

- Architected scalable infrastructure for an LLM fine-tuning platform including staging environments, model hub, deployment pipelines, and experiment management.
- Developed a fine-tuning framework supporting LoRA, QLoRA, Supervised Fine-Tuning (SFT), Direct Preference Optimization (DPO), Parameter Efficient Fine-tuning (PEFT), and Full Fine-Tuning — directly applicable to fine-tuning and adapting speech and language models.

- Built automated MLOps pipelines for dataset management, experiment tracking, model versioning, and deployment workflows.
- Developed model optimization pipelines including quantization and runtime conversion using llama.cpp, ONNX, and TensorRT to improve inference speed for production deployment.
- Implemented evaluation pipelines for RAG applications including hallucination detection, answer relevance, and context scoring — built offline evaluation and regression-testing methodology for model releases.
- Integrated Prometheus, Grafana, cAdvisor, and Node Exporter to enable production monitoring and system observability.

GenAI Developer Intern

Mar 2025 – Jul 2025

CloudAngles, Hyderabad

- Developed a centralized system monitoring dashboard for internal AI products.
- Integrated customized computer vision models with Roboflow for dynamic image classification.
- Built AI agents using LangGraph to automate enterprise workflows.
- Developed reusable UI components for chatbot and AI agent creation platforms.

SDE Intern

Dec 2024 – Jan 2025

Prathibha Innovations, Hyderabad (Remote)

- Developed a custom user management system using Express.js.
- Configured Prisma ORM for scalable and maintainable database operations.

Education

Keshav Memorial Engineering College

2021 – 2025

Bachelor of Engineering in Computer Science and Engineering (**Specialization: AI & ML**)

Osmania University, Hyderabad

CGPA: 8.14

Relevant Coursework: Machine Learning, Advanced Machine Learning, Deep Learning, Operating Systems, Computer Networks, Database Management Systems, Data Structures and Algorithms, Image Processing, Information Retrieval.

Projects

In-House LLM Summarization Application

- Developed an enterprise text summarization application using DistilBART.
- Fine-tuned transformer models on custom datasets for improved summarization quality.
- Built an automated document summarization workflow for enterprise knowledge management.

Employee Attrition Predictor

- Developed a machine learning web application for predicting employee attrition using Flask.
- Applied feature engineering and feature selection techniques to improve prediction accuracy.
- Built an end-to-end prediction pipeline providing HR teams with actionable employee retention insights.

Achievements

Gold Medal

Awarded Gold Medal in the State Level Science Talent Search Examination (SLSTE), demonstrating excellence in science and analytical problem solving.